PROJECT REPORT

Fake Review Detection using Text Frequency

Student Name	Asna.M
Register Number	2117250020041
Faculty Guide	Dr. Anwar Basha
Department	Computer Science and Engineering

1. Introduction

Fake reviews are false or misleading opinions shared on online platforms to influence customer decisions. These reviews may either promote a product unfairly or damage the reputation of competitors. As online shopping and digital marketing continue to grow, fake reviews have become a major issue for customers and businesses. This project aims to identify fake reviews by analyzing text frequency patterns and repeated keywords within review content. The system helps improve trust and transparency in online review platforms.

2. Problem Statement

Online shopping websites receive thousands of reviews every day, making it difficult to manually identify fake reviews. Many businesses use manipulated reviews to attract customers or harm competitors. Traditional review systems often fail to detect suspicious content effectively. This creates confusion among customers and reduces trust in online platforms. Therefore, an automated fake review detection system is required to analyze review text accurately and efficiently.

3. Objectives

The main objective of this project is to detect fake reviews using text frequency analysis techniques. The project also aims to preprocess review text, remove duplicate words, identify unusual keyword repetitions, and classify reviews as genuine or fake. Another important objective is to improve customer confidence by ensuring that online reviews are reliable and trustworthy.

4. Methodology

The system first collects review data from various online platforms. The collected reviews are cleaned through preprocessing techniques such as tokenization, stop-word removal, and stemming. Text frequency analysis is then applied to identify repeated patterns and suspicious keywords. Machine learning techniques may also be used to classify reviews more accurately. Finally, the system displays whether the review is fake or genuine based on the analysis results.

5. Tools and Technologies

Python is used as the primary programming language for implementing the project. Natural Language Processing (NLP) techniques are used for analyzing textual review data. Libraries such as NLTK, Pandas, and Scikit-learn are used for preprocessing, feature extraction, and classification. Datasets collected from e-commerce websites are used for training and testing the system.

6. Advantages

The proposed system improves customer trust by identifying misleading reviews effectively. It reduces the impact of spam reviews and helps customers make better purchasing decisions. The system is cost-effective, fast, and capable of analyzing large datasets with improved accuracy. It also supports businesses in maintaining genuine customer feedback.

7. Applications

Fake review detection systems are widely used in e-commerce websites, restaurant review platforms, hotel booking applications, and online recommendation systems. The system can also be used in social media platforms and product feedback analysis systems to identify suspicious user-generated content.

8. System Architecture

The architecture of the proposed system consists of data collection, preprocessing, feature extraction, text frequency analysis, classification, and result generation modules. Each module works together to improve fake review detection performance and ensure accurate classification.

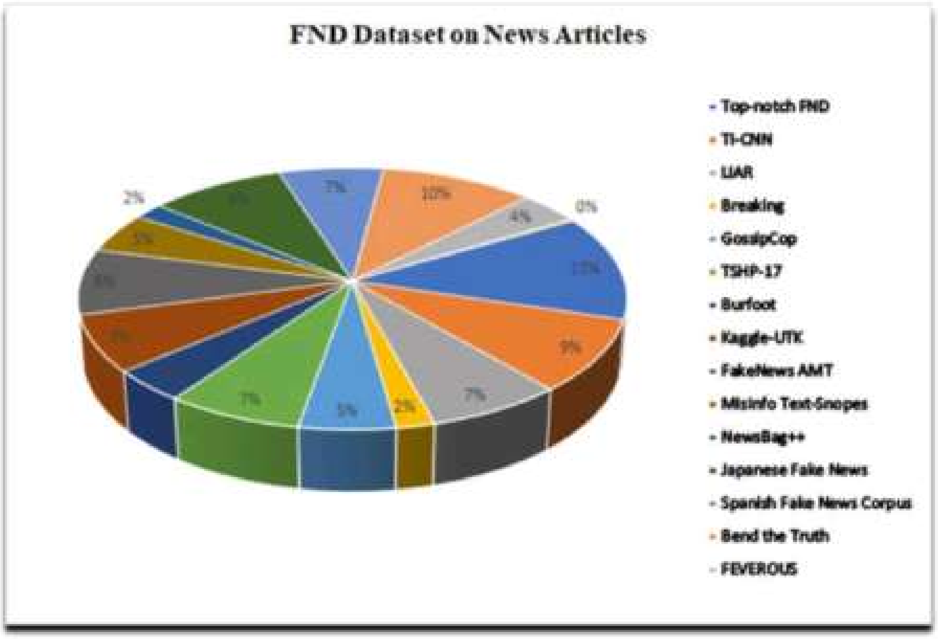
9. Expected Output

The expected output of the project is the automatic identification of fake and genuine reviews. The system generates a report showing suspicious review patterns, repeated keywords, and final classification results. This helps users understand whether a review can be trusted or not.

10. Conclusion and Future Scope

Fake Review Detection using Text Frequency provides an effective solution for reducing misleading content in online platforms. The project improves transparency and helps customers make informed decisions. In the future, advanced deep learning and artificial intelligence techniques can be integrated to improve detection accuracy and handle larger datasets more efficiently.

pie chart



Bar graph